

Raunaq Singh

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. Industrial Engineering — Concentration: AI/OR for Decisions | Minor: Computing and Theory

Transfer student (Fall 2026), previously University of Illinois Urbana-Champaign | Expected Graduation: May 2029

Relevant Coursework: Linear Algebra, Discrete Mathematics, Multivariable Calculus, Probability with Applications, Intro to Computing, Object-Oriented Programming, Financial Accounting

TECHNICAL SKILLS

Quant / Finance: Quantitative Research, Statistical Arbitrage, Cointegration Testing, Backtesting, Multiple-Testing Correction, Risk Management, Financial Modeling

ML / Data: scikit-learn, pandas, NumPy, statsmodels, XGBoost, Time Series Analysis, NLP, Feature Engineering, Exploratory Data Analysis

Languages: Python, SQL, TypeScript, JavaScript

Software / Tools: Node.js, Express, MongoDB, React, Next.js, Astro, REST APIs, Git, Jest, matplotlib, Power BI, Jupyter, yfinance

EXPERIENCE

Software Engineering Intern — MiraPhoto

2026

Early-stage startup engineering team (Node.js / Express / MongoDB backend; Astro / React frontend)

- Independently owned a production-bound Organization Profile & Event RSVP feature end-to-end, reverse-engineering the existing UI design system (layout, typography, styling) from a single screenshot with no source access.
- Migrated the feature from Next.js to Astro to meet static-frontend requirements, resolving real routing, CI/CD (GitHub Actions, Node version pinning), and hosting constraints across GitHub Pages and Vercel.
- Integrated the Astro frontend with the live Node.js/Express/MongoDB backend by directly reading route handlers, Zod schemas, and Swagger docs; verified end-to-end against production and staging, independently diagnosing and resolving an undocumented schema mismatch with the backend engineer.
- Implemented role-based access control (organization/photographer/admin) and strict form validation (E.164 phone, URL, email, length constraints) using React Hook Form + Zod; scoped an ambiguous organization-account requirement into a transparent placeholder flow rather than a misleading one.

TECHNICAL PROJECTS

Mean-Reversion Statistical Arbitrage System

Python, pandas, NumPy, statsmodels, yfinance

- Built an end-to-end, leakage-free pairs-trading research pipeline: economically-motivated candidate selection, log-price cointegration testing, and Benjamini-Hochberg FDR correction to control false discoveries across simultaneous significance tests.
- Designed train/test-separated hedge-ratio estimation and a vectorized z-score signal generation and backtesting engine incorporating transaction costs and stop-loss controls, with all performance metrics computed strictly out-of-sample.
- Identified one statistically significant cointegrated pair (KO-PEP, $p = 0.036$ post-correction) from 8 candidates; ran a full diagnostic suite (transaction-cost sensitivity, statistical-power check, walk-forward stability) that surfaced no robust tradeable edge and closed the project as a rigorously documented negative result rather than overfitting a fix.

AirBnB Pricing Analysis: NYC vs. Chicago Market Comparison

Python, pandas, NumPy, scikit-learn, seaborn, matplotlib

- Cleaned and analyzed AirBnB listing data across New York City and Chicago (Kaggle open datasets), removing outliers and building comparative visualizations (price distributions by room type, neighborhood-level heatmaps); found NYC priced modestly higher (\$106 vs. \$99 median), with premium listings clustered in Manhattan/Brooklyn versus Chicago's Loop and Near North Side.
- Built linear regression and KMeans clustering models to predict price from listing features (reviews, availability, host activity); both showed low explanatory power (R^2 of 0.13 and 0.10), an honest result showing surface-level features alone don't fully explain AirBnB pricing.

RESEARCH

BU1LD Research Program — Selected Participant

Financial Machine Learning & Quantitative Finance Team

- Selected from an international applicant pool for a research program pairing students with faculty-adjacent researchers; assigned to a team applying ML methods to quantitative finance and market forecasting.
- Ran a diagnostic pass (transaction-cost sensitivity, statistical-power, walk-forward stability) on the team's pairs-trading strategy, concluded no robust signal, and pre-registered a v2 protocol (intraday crypto pairs, cointegration z-scores, 6-fold walk-forward, explicit pass/fail criteria) to avoid overfitting.

LEADERSHIP & ACTIVITIES

- Founder, Boro Bike Club — Launched and organized a student club from the ground up, managing logistics, recruitment, and community partnerships.
- Co-Founder, Community Swim — Co-founded a community swim initiative, coordinating programming and operations.
- Captain, Varsity Swim Team — Led a competitive athletics team, mentoring teammates and managing team accountability and performance standards.